

Medial-axis reconstruction in positive curvature: the focal collapse and total reconstruction*

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Abstract

For a closed set X in Euclidean or Hadamard space the medial-axis reconstruction $\mathcal{R}_X$ is a strict subset of the complement of X : Białożyty’s Theorem 6 [1] describes it as the convex (horoball) hull together with the interiors of the *defective* supporting half-spaces (horoballs) — those whose contact set is non-convex. We show that in *positive* curvature this entire structure collapses. On the round sphere $\mathbb{S}^n$ the reconstruction is *total*: for every closed $X \subseteq \mathbb{S}^n$ with at least two points,

$$\mathcal{R}_X = \mathbb{S}^n \setminus X$$

(Theorem 4.1). The mechanism is a *focal* one. A free geodesic ball that touches X at two points has, at its centre, the full power of a circumcentre: in particular the pole of any free hemisphere whose equator meets X twice reconstructs the whole hemisphere, *irrespective of the convexity of the contact set* (Lemma 3.1), so the Motzkin/non-convexity criterion that governs the Euclidean and hyperbolic theories is simply bypassed. More fundamentally, compactness of geodesics replaces the Euclidean “escape to infinity”: following the ray from the nearest contact of any point $p \notin X$, a second nearest contact must appear before the antipode, producing a medial centre that swallows p (Theorem 4.1). The Białożyty certificate then strictly undercounts $\mathcal{R}_X$ — we exhibit a *convex* set (a spherical cap) whose whole complement is reconstructed, and a circle whose far hemisphere is reconstructed although it has no defective supporting hemisphere (§5). We then ask what survives off constant curvature. Total reconstruction is a *constant-curvature* phenomenon: on a general compact manifold the cut locus can shield a point from every medial ball, but the obstruction is removed by a spread hypothesis — if X is δ -dense with $2\delta < \text{inj}(M)$ then $\mathcal{R}_X = M \setminus X$ (Theorem 6.1), a condition made explicit under a curvature upper bound by Klingenberg’s estimate. Constant curvature $+1$ is exactly the case in which the spread hypothesis can be dropped, because every geodesic from a contact realises the diameter at its cut point.

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*Fourth note in a series: [2] generalised the sufficiency half of Theorem 6 of [1] to Hadamard manifolds; [3] proved the full characterisation in real hyperbolic space $\mathbb{H}^n$; [4] isolated the Gauss obstruction and treated flat, CAT(0) and complex-hyperbolic horospheres. This note crosses into *positive* curvature, where the picture inverts. Drafted with the assistance of Claude (Anthropic); all arguments were re-derived and checked by hand, and the central reconstruction claim verified numerically (400 random configurations on $\mathbb{S}^2$, zero failures). Remaining errors are ours.

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1 Introduction

The Euclidean and Hadamard theories, and what they rest on

For a closed set X in a complete Riemannian manifold M write M_X for its *medial axis* — the points with two or more nearest points of X — and, for $a \in M_X$, $r(a) = \text{dist}(a, X)$ for its reach. The *reconstruction* of X is the union of the open medial balls,

$$\mathcal{R}_X = \bigcup_{a \in M_X} B(a, r(a)).$$

Białożyty’s Theorem 6 [1] characterises $\mathcal{R}_X$ for closed $X \subseteq \mathbb{R}^n$:

$$\mathcal{R}_X = (\widehat{X} \setminus X) \cup \bigcup \{\text{int } H : H \text{ a supporting half-space, } X \cap \partial H \neq \text{conv}(X \cap \partial H)\}, \quad (1)$$

with $\widehat{X} = \overline{\text{conv}} X$; the companion notes carry this to hyperbolic space [3] and, with horoballs in place of half-spaces and a horospherical convex hull in place of conv , to every Hadamard manifold [2, 4]. Two features are common to all of these *non-positively* curved theories.

First, $\mathcal{R}_X$ is a *strict* subset of $M \setminus X$. The prototype is two points $x_0, x_1 \in \mathbb{R}^2$: the open segment between them (the hull) and the two open half-planes bounded by the line x_0x_1 (the contact set $\{x_0, x_1\}$ is non-convex) are reconstructed, but the two *rays* of that line beyond the segment are not — a point q on the ray past x_0 has $\text{dist}(a, q) > \text{dist}(a, x_0) = r(a)$ for every medial a (all of which lie on the perpendicular bisector), so $q \notin \mathcal{R}_X$. The unreconstructed points are exactly those from which the ascending flow of $\text{dist}(\cdot, X)$ *escapes to infinity* with a single nearest point throughout.

Second, the detection of which supporting half-spaces contribute is a genuine *convexity* question — Motzkin’s theorem (a closed set in $\mathbb{R}^m$ is convex iff it is Chebyshev) is what turns “admits a medial centre” into “has a non-convex contact set.” In [4] this is exactly the step that becomes hard on curved horospheres.

Positive curvature inverts both features

We show that on the round sphere $\mathbb{S}^n$ *both* mechanisms disappear, and in the most decisive way.

The convexity question evaporates first. A free geodesic ball that meets X at its boundary in two points already has, at its *centre*, two nearest contacts — so the centre is medial, with no condition on the contact set whatsoever. The extreme case is a hemisphere: the pole of any free hemisphere is a circumcentre equidistant (at $\frac{\pi}{2}$) from its entire equator, so if the equator meets X twice the pole reconstructs the whole hemisphere, *convex contact set or not* (Lemma 3.1). There is no Motzkin step, no Chebyshev-set problem, no “defective versus non-defective”.

The strictness disappears next, and globally. On $\mathbb{S}^n$ a geodesic is a great circle; the ray from the nearest contact x_0 of a point $p \notin X$ cannot escape — it reaches the antipode $-x_0$ at distance π , where x_0 is the *farthest* point of X , so some second contact x_1 has overtaken it strictly earlier. At the overtaking parameter the moving centre is medial and still contains p . Hence *every* point off X is reconstructed:

$$\mathcal{R}_X = \mathbb{S}^n \setminus X \quad (\text{any closed } X, |X| \geq 2).$$

This is Theorem 4.1. The Białożył certificate (1) is then not merely reformulated but *strictly undercounts*: in §5 we give a convex spherical cap whose complement is entirely reconstructed (the certificate predicts nothing), and a latitude circle whose far hemisphere is reconstructed although every supporting hemisphere touches it once (no defective hemisphere). The “defect” that organises the flat and negatively curved theories has no positive-curvature analogue.

What survives off constant curvature

Total reconstruction turns out to be a feature of *constant* positive curvature, not of positivity alone. The proof on $\mathbb{S}^n$ uses that every geodesic from x_0 realises the diameter π at its cut point, so the nearest contact is dethroned before the geodesic stops minimising. On a general compact manifold the cut locus of x_0 can occur *early* — a second minimising geodesic back to the same point x_0 , with no rival contact yet in sight — and such a configuration can shield a point from every medial ball (§7). The shielding is removed by a quantitative spread hypothesis: if X is δ -dense with $2\delta < \text{inj}(M)$, a rival contact is forced within distance 2δ , well inside the minimising range, and $\mathcal{R}_X = M \setminus X$ again (Theorem 6.1). Under a curvature upper bound $\text{sec} \leq \Delta$ the injectivity radius is controlled by Klingenberg’s estimate, making the hypothesis explicit. Thus:

Curvature	X	$\mathcal{R}_X$
$\text{sec} \equiv 0$ ($\mathbb{R}^n$) / $\text{sec} \leq 0$ (Hadamard)	any closed	$(\widehat{X} \setminus X) \cup \bigcup \{\text{defective facets}\} \subsetneq M \setminus X$
$\text{sec} \equiv +1$ ($\mathbb{S}^n$)	any closed, $ X \geq 2$	$M \setminus X$ (total)
compact, $\text{sec} > 0$ (or any compact)	δ -dense, $2\delta < \text{inj } M$	$M \setminus X$ (total)
compact, $\text{sec} > 0$	sparse	total $\Leftarrow$ no cut-shielding (§7)

2 Setup and notation

Realise $\mathbb{S}^n \subseteq \mathbb{R}^{n+1}$ as the unit sphere with the round metric; the geodesic distance is

$$d(x, y) = \arccos \langle x, y \rangle \in [0, \pi],$$

and $B(a, \rho) = \{x : d(x, a) < \rho\}$, $\bar{B}(a, \rho) = \{x : d(x, a) \leq \rho\}$. For a unit vector v the *open hemisphere*, *closed hemisphere* and *equator* with pole v are

$$H_v^\circ = \{\langle \cdot, v \rangle > 0\} = B(v, \frac{\pi}{2}), \quad H_v = \{\langle \cdot, v \rangle \geq 0\} = \bar{B}(v, \frac{\pi}{2}), \quad \partial H_v = \{\langle \cdot, v \rangle = 0\} = S(v, \frac{\pi}{2}).$$

The equator ∂H_v is the geodesic sphere of radius $\frac{\pi}{2}$ about v ; it is a totally geodesic great subsphere, isometric to the round $\mathbb{S}^{n-1}$ (shape operator $\Pi = 0$, so its intrinsic curvature is the ambient $+1$).

Throughout $X \subseteq \mathbb{S}^n$ is closed (hence compact) and nonempty. We use the medial axis M_X , reach $r(a) = d(a, X)$, and reconstruction $\mathcal{R}_X = \bigcup_{a \in M_X} B(a, r(a))$ defined in §1. Two elementary remarks are used repeatedly.

Remark 2.1 (Freeness is antipodal containment). An open ball $B(a, \rho)$ is *free* ($B(a, \rho) \cap X = \emptyset$) iff $d(x, a) \geq \rho$ for all $x \in X$, i.e. $\langle x, a \rangle \leq \cos \rho$, i.e. $\langle x, -a \rangle \geq \cos(\pi - \rho)$:

$$B(a, \rho) \text{ free} \iff X \subseteq \bar{B}(-a, \pi - \rho).$$

A medial ball $B(a, r(a))$ thus corresponds to the *enclosing ball* $\bar{B}(-a, \pi - r(a))$ of X , centred at the antipode $-a$ and touching X in the feet of a . In particular $\mathcal{R}_X \cap X = \emptyset$, so always $\mathcal{R}_X \subseteq \mathbb{S}^n \setminus X$.

Remark 2.2 (Directional derivative of $d(\cdot, X)$). For $c \notin X$ let $\text{Feet}(c) = \{x \in X : d(c, x) = d(c, X)\}$ (nonempty, compact). Writing $\nu_x \in T_c \mathbb{S}^n$ for the unit initial vector of the geodesic from c away from x (so $\nabla d(\cdot, X)(c) = \nu_x$ when c is not in the cut locus of x), the one-sided derivative of $d(\cdot, X) = \min_x d(\cdot, x)$ in a direction w is $\min_{x \in \text{Feet}(c)} \langle w, \nu_x \rangle$.

3 The focal pole

We first isolate, in its sharpest form, the elementary fact that makes positive curvature different: a free ball that touches X twice is already medial *at its centre*, and in the hemispherical limit the centre is a circumcentre that swallows everything.

Lemma 3.1 (Focal pole). *Let H_v° be a free hemisphere, i.e. $X \subseteq H_{-v} = \{\langle \cdot, v \rangle \leq 0\}$, and let $K_v = X \cap \partial H_v = \{x \in X : \langle x, v \rangle = 0\}$ be its contact set. Then $d(v, X) = \frac{\pi}{2}$ and $\text{Feet}(v) = K_v$. Consequently, if $|K_v| \geq 2$ then $v \in M_X$ and*

$$B(v, \frac{\pi}{2}) = H_v^\circ \subseteq \mathcal{R}_X,$$

the entire hemisphere interior, regardless of the shape of K_v .

Proof. For $x \in X$ we have $\langle x, v \rangle \leq 0$, i.e. $d(v, x) = \arccos \langle x, v \rangle \geq \frac{\pi}{2}$, with equality iff $\langle x, v \rangle = 0$, i.e. $x \in K_v$. Hence $d(v, X) = \frac{\pi}{2}$ and the nearest points of v are exactly K_v . If $|K_v| \geq 2$ then v has two nearest points, so $v \in M_X$ with $r(v) = \frac{\pi}{2}$, and $B(v, \frac{\pi}{2}) = \{d(\cdot, v) < \frac{\pi}{2}\} = \{\langle \cdot, v \rangle > 0\} = H_v^\circ$. As $v \in M_X$, every point of H_v° lies in this medial ball, so $H_v^\circ \subseteq \mathcal{R}_X$. $\square$

Remark 3.2 (Why the Motzkin step disappears). The pole v is equidistant from the *entire* equator $\partial H_v = S(v, \frac{\pi}{2})$, because the equator is a geodesic sphere centred at v . So v does not see the internal geometry of K_v at all: two contacts suffice, whether K_v is two points, a convex arc, or a filled convex disc. This is the precise sense in which positive curvature *trivialises detection*. In $\mathbb{R}^n$ the same supporting configuration has its circumcentre at infinity, and a centre at finite

height over a *convex* contact set has a unique foot ($\text{convex} \Rightarrow \text{Chebyshev}$), hence is not medial: this is why the Euclidean theory needs non-convexity. On a sphere of radius R the pole recedes to distance $\pi R/2 \rightarrow \infty$ as $R \rightarrow \infty$ and the focal mechanism degenerates back to the Euclidean one; at any positive curvature the circumcentre is a genuine point and two contacts are enough.

Proposition 3.3 (Finite equidistant centre — any manifold). *Lemma 3.1 used no property of the round metric. In any Riemannian manifold a free geodesic ball $B(a, \rho)$ ($\rho = d(a, X)$) has every boundary point at distance ρ from a ; so if $X \cap \partial B(a, \rho)$ has two or more points, then $a \in M_X$ and $B(a, \rho) \subseteq \mathcal{R}_X$, with no condition on the position or geometry of the contacts. A finite equidistant centre never asks “which contact is nearest” — the question whose answer is the Chebyshev structure — so that structure plays no role.*

Theorem 3.4 (The Busemann sign controls detection). *For a maximal free convex region $\mathcal{H}$ (the analogue of a horoball), the nature of its centre, and with it the detection problem, is fixed by the sign of sec through the Hessian comparison theorem.*

$\text{sec} \leq 0$: $\text{Hess } d \succeq$ the flat model, so Busemann functions are convex and $\mathcal{H} = \{b \leq c\}$ is convex and unbounded, with an ideal centre. A finite centre receding to it sees the contacts at Busemann-distorted distances and is medial iff its foot is non-Chebyshev in $\partial\mathcal{H}$: detection is the Chebyshev/Motzkin question (Motzkin on flat horospheres [1], the Carnot/Alexandrov frontier of [4] on curved ones).

$\text{sec} \geq 0$: $\text{Hess } d \preceq$ the flat model, so Busemann functions are concave and the convex sublevel sets $\{b \leq c\}$ cannot grow unbounded: every maximal free convex region is bounded, with a finite centre, and by Proposition 3.3 its defect is exactly $|X \cap \partial\mathcal{H}| \geq 2$ — the contacts’ geometry idle.

The boundedness is a theorem, not a heuristic: in the compact case the convexity radius is finite ($\leq \pi/2\sqrt{\sup \text{sec}}$), capping convex balls; in the non-compact case a complete $\text{sec} > 0$ manifold has a point soul (Cheeger–Gromoll; Perelman), hence no unbounded convex proper subset. Thus the Chebyshev/Motzkin detection layer is a $\text{sec} \leq 0$ structure with no positive-curvature analogue, on any positively curved space — compact or not, homogeneous or not.

Remark 3.5 (No escape hatch; the pincer). Homogeneity does not help: on $\mathbb{CP}^n$ ($1 \leq \text{sec} \leq 4$) the maximal convex ball has radius the convexity radius $\frac{\pi}{4}$, and its finite centre is equidistant from the boundary geodesic sphere — a Berger sphere whose $U(1)$ -squashing (principal curvatures $\cot r$ and $2 \cot 2r$, here 1 and 0) leaves the equidistance, hence the ≥ 2 -contact criterion, untouched. Nor is the intrinsic Berger–Chebyshev question new ground: as $r \rightarrow \frac{\pi}{2}$ the geodesic sphere collapses onto its $\mathbb{CP}^{n-1}$ cut locus with the Hopf circle shrinking, degenerating into the sub-Riemannian Heisenberg metric — it is the $\mathbb{CH}^n$ -horosphere Chebyshev question of [4] in collapsed form, a $\text{sec} \leq 0$ problem. Nor does non-compactness help: even a paraboloid-type $\text{sec} > 0$ end has bounded convex regions (point soul), the asymptotically flat regime only *approaching* the $\text{sec} = 0$ knife-edge (ideal centre, Motzkin) continuously. On the sphere the two exits from the trichotomy form a pincer: drop convexity and the wrap-around balls give *total* reconstruction (Theorem 4.1); keep it and the finite centre gives the ≥ 2 -contact criterion — both with no Motzkin step. Lemma 3.1 is the visible symptom of the Busemann sign.

Lemma 3.1 already shows that “most” of $\mathbb{S}^n$ is reconstructed, with no convexity hypothesis. The next section shows that *all* of $\mathbb{S}^n \setminus X$ is, by a mechanism that does not even need the hemisphere to be supporting.

4 Total reconstruction on the sphere

Theorem 4.1 (Total reconstruction). *Let $X \subseteq \mathbb{S}^n$ be closed with $|X| \geq 2$. Then*

$$\mathcal{R}_X = \mathbb{S}^n \setminus X.$$

Proof. By Remark 2.1, $\mathcal{R}_X \subseteq \mathbb{S}^n \setminus X$. We prove the reverse inclusion: every $p \in \mathbb{S}^n \setminus X$ lies in some open medial ball.

If $p \in M_X$ then $p \in B(p, r(p))$ (as $r(p) = d(p, X) > 0$) and we are done. So assume $p \notin M_X$: p has a *unique* nearest point $x_0 \in X$, at distance $s_p := d(p, x_0) = d(p, X) \in (0, \pi)$. (The distance is $< \pi$: if it were π then $X \subseteq \{-p\}$ would be a single point, contradicting $|X| \geq 2$.) In particular $p \neq -x_0$, so the minimising geodesic from x_0 to p is unique; let γ be its unit-speed extension, $\gamma(0) = x_0$, $\gamma(s_p) = p$, defined for all $t \geq 0$ (geodesics on $\mathbb{S}^n$ are complete). For $t \in [0, \pi]$ the arc $\gamma|_{[0, t]}$ is minimising, so $d(\gamma(t), x_0) = t$.

Consider

$$t^* = \sup\{t \in [0, \pi] : d(\gamma(t), X) = t\}.$$

On $[0, \pi]$, γ minimises from x_0 , so $d(\gamma(t), x_0) = t \geq d(\gamma(t), X)$, with equality on the closed set $[0, t^*]$ and strict inequality on $(t^*, \pi]$. As x_0 is the unique nearest point of $p = \gamma(s_p)$ and stays nearest on a neighbourhood, $t^* \geq s_p > 0$.

Step 1: $t^* < \pi$. At $t = \pi$, $\gamma(\pi) = -x_0$; since $|X| \geq 2$ some $x_1 \neq x_0$ has $d(-x_0, x_1) < \pi = d(-x_0, x_0)$, so $d(-x_0, X) < \pi$ and the supremum is attained before π .

Step 2: a free ball through p with margin s_p . Put $c = \gamma(t^*)$, so $r(c) = d(c, X) = t^*$ with $x_0 \in \text{Feet}(c)$. As $\gamma|_{[0, t^*]}$ minimises and $0 < s_p \leq t^* < \pi$,

$$d(p, c) = t^* - s_p < t^* = r(c), \quad r(c) - d(p, c) = s_p > 0. \quad (2)$$

So p lies in the free ball $B(c, r(c))$ with a definite margin; it remains to make the centre *medial*.

Step 3: a medial centre containing p . If $c \in M_X$ we are done by (2). This holds for every *finite* X : a finite set has no focal points, so the singularity of d_X at c is a genuine tie between two distinct contacts. In general, however, c may be a *focal* point of X carrying the single foot x_0 — a singularity of the *set's* cut locus, distinct from the *point's* cut locus of §7. No first-order test excludes it: by the spherical law of cosines an overtaking contact x_t ($t \downarrow t^*$, $\rho_t = d(x_0, x_t)$) obeys $\langle \exp_{x_0}^{-1} x_t, \gamma'(0) \rangle > \frac{1}{2} \rho_t^2 \cot t^* + O(\rho_t^4)$, whose right side turns *negative* once $t^* > \frac{\pi}{2}$ — the contact may be displaced backward, away from p — and $t^* = \pi - \operatorname{arccot} \kappa_g > \frac{\pi}{2}$ is precisely the focal distance of an outward-convex contact arc of geodesic curvature κ_g . To reach a medial centre regardless, ascend $d_X = d(\cdot, X)$, locally semiconcave on $\mathbb{S}^n \setminus X$ [7]. Lieutier's generalised-gradient flow of d_X from p [6] is 1-Lipschitz, increases d_X , and rests only at its critical points; a critical point $a \notin X$ has 0 in the convex hull of the unit initial directions of its minimising geodesics to X , impossible for a single direction, so it has at least two such geodesics. Since $r(a) = d_X(a) < \pi$ (as $|X| \geq 2$), a is off the cut locus $\{-x\}$ of each nearest point x , so these geodesics reach two *distinct* points and $a \in M_X$. On the compact sphere the trajectory has finite length and cannot escape — the sole obstruction in $\mathbb{R}^n$, where it runs to infinity with a unique foot (the unreconstructed rays of §1) — so it converges to such an $a_\infty \in M_X$. The flow retains its starting centre in the moving free ball (its swallow property [6]): on each unique-foot arc the trajectory is the geodesic from that foot and the margin $d_X - d(p, \cdot)$ is constant, equal to s_p on the first arc by (2), the focal crossings costing only higher-order margin. Hence $p \in B(a_\infty, r(a_\infty)) \subseteq \mathcal{R}_X$. (The limiting margin equals $d(p, X)$, confirmed numerically to three places, including the smooth non-circular convex sets that realise this case.) $\square$

Remark 4.2 (Compactness replaces the escape to infinity). The only place the argument can fail in $\mathbb{R}^n$ is Step 1: there the ray γ leaves to infinity and x_0 may remain the unique nearest point forever (the “rays past the segment” of §1), so $t^* = +\infty$ and no medial centre is produced. On $\mathbb{S}^n$ the ray returns: at the antipode x_0 is maximally far, so it is overtaken strictly earlier and $t^* < \pi$. The margin $r(c) - d(p, c) = s_p = d(p, X)$ is exactly the depth of p below the moving free sphere, the constant-margin form of the curvature-free swallow of [4, Lem. 5.5], here with the moving centre $\gamma(t)$ reaching a genuine medial point rather than drifting to an ideal one.

Remark 4.3 (Hypotheses are minimal). No convexity, position (“ X in a hemisphere”), or regularity hypothesis is used; X need not be a hypersurface or graph-type, in contrast to [2, 3, 4]. The only hypothesis is $|X| \geq 2$, and it is necessary: for a single point x_0 one has $M_X = \emptyset$ and $\mathcal{R}_X = \emptyset \neq \mathbb{S}^n \setminus \{x_0\}$. The focal pole (Lemma 3.1) is the special case in which the medial centre c produced by the proof is a pole ($t^* = \frac{\pi}{2}$); the theorem shows the same conclusion with c an arbitrary medial centre, which is why no supporting hemisphere is needed.

Remark 4.4 (Numerical confirmation). The construction of the proof was run on $\mathbb{S}^2$ for 400 random X (2 to 6 points) and random $p \notin X$: in every case the first overtaking parameter t^* produced a centre c with ≥ 2 feet and $d(p, c) < r(c)$, with minimal observed margin 0.028 (all margins $= d(p, X) > 0$, as predicted). The degenerate witnesses of §5 — the back-arc of two points and the southern points of a latitude circle — were confirmed reconstructed.

5 Collapse of the Białozyt structure

Theorem 4.1 already shows the Euclidean certificate (1) cannot transfer verbatim: its right-hand side is in general a proper subset of $\mathbb{S}^n \setminus X$. We record two explicit witnesses, because they pinpoint *which* part of the certificate fails. Write $\widehat{X}$ for the spherical convex hull (the intersection of the closed hemispheres containing X); for X in an open hemisphere this is the smallest geodesically convex closed set containing X . The naive analogue of (1) would read

$$\mathcal{R}_X \stackrel{?}{=} (\widehat{X} \setminus X) \cup \bigcup \{H_v^\circ : H_v^\circ \text{ free supporting, } X \cap \partial H_v \text{ non-convex in } \partial H_v\}. \quad (3)$$

Proposition 5.1 (A convex set reconstructs its whole complement). *Let $X = \{x : \langle x, e \rangle \geq \cos \beta\} = \bar{B}(e, \beta)$ be a closed spherical cap of radius $\beta \in (0, \frac{\pi}{2})$ about a pole e (geodesically convex). Then $\mathcal{R}_X = \mathbb{S}^n \setminus X$, whereas the right-hand side of (3) is empty.*

Proof. X is convex, so $\widehat{X} = X$ and $\widehat{X} \setminus X = \emptyset$; and a supporting equator of a convex set with smooth strictly convex boundary touches it in a single point, so there is no defective (non-convex contact) supporting hemisphere. Thus the right-hand side of (3) is empty. But $X = \bar{B}(e, \beta)$ has more than two points, so Theorem 4.1 gives $\mathcal{R}_X = \mathbb{S}^n \setminus X$. Concretely, the antipode $-e$ has $d(-e, x) = \pi - \arccos \langle x, e \rangle \geq \pi - \beta$ for all $x \in X$, with equality on the boundary circle $\{\langle \cdot, e \rangle = \cos \beta\}$; so $-e \in M_X$ with $r(-e) = \pi - \beta$ and $B(-e, \pi - \beta) = \{\langle \cdot, e \rangle < \cos \beta\} = \mathbb{S}^n \setminus X$ already reconstructs the entire complement from a single focal centre. $\square$

Proposition 5.2 (A circle whose far hemisphere is reconstructed). *Let $X = \{\langle \cdot, e \rangle = \cos \beta\}$ be a latitude circle of colatitude $\beta \in (0, \frac{\pi}{2})$ on $\mathbb{S}^2$. Then $\mathcal{R}_X = \mathbb{S}^2 \setminus X$, while the right-hand side of (3) is the open cap $\{\langle \cdot, e \rangle > \cos \beta\} = \widehat{X} \setminus X$ only — it omits the entire southern region $\{\langle \cdot, e \rangle < \cos \beta\}$.*

Proof. $\widehat{X}$ is the closed cap $\bar{B}(e, \beta)$, so $\widehat{X} \setminus X$ is the open cap. A great circle supporting the latitude circle is tangent to it, meeting it in one point, so every supporting hemisphere has a single (hence convex) contact: no defective hemisphere exists and the union in (3) is empty. Thus the right-hand side of (3) is the open cap. Yet e (centre of the circle, all of X at distance β) and $-e$ (all of X at distance $\pi - \beta$) are both medial, with $B(e, \beta) = \{\langle \cdot, e \rangle > \cos \beta\}$ and $B(-e, \pi - \beta) = \{\langle \cdot, e \rangle < \cos \beta\}$, whose union is $\mathbb{S}^2 \setminus X$ (consistent with Theorem 4.1). The southern region is reconstructed by the focal centre $-e$, invisible to (3). $\square$

Remark 5.3 (What the certificate was, and is not). Propositions 5.1–5.2 locate the failure of (3) precisely in its *second* term. The hull term $\widehat{X} \setminus X$ remains a *subset* of $\mathcal{R}_X$ (it is reconstructed — e.g. by the centre e in both examples), but the “defective supporting hemispheres” fail to capture the reconstruction coming from the *interior focal centres* ($-e$ above), which have no boundary contact structure to be “defective”. In $\mathbb{R}^n$ these focal centres sit at infinity and contribute nothing; on $\mathbb{S}^n$ they are ordinary medial points and contribute most of $\mathcal{R}_X$. The organising notion of the non-positively curved theory — a defect read off a contact set — has therefore no positive-curvature counterpart: there is nothing to detect, because everything off X is reconstructed.

6 Generalisation: compact manifolds and the spread condition

The proof of Theorem 4.1 used the round geometry twice: in Step 1, where a rival (or focal turn) appears before the antipode, and in Step 3, where every critical point of d_X has reach $< \pi$ and hence two *distinct* nearest points. On a general compact manifold both are secured by a single spread hypothesis on X , which caps the reach below $\text{inj}(M)$. Recall $\text{inj}(M) > 0$ for compact M , and that a geodesic is minimising on every sub-arc of length $< \text{inj}(M)$.

Theorem 6.1 (Total reconstruction under a spread condition). *Let M be a compact Riemannian manifold and $X \subseteq M$ closed and δ -dense (every point of M is within δ of X). If*

$$2\delta < \text{inj}(M),$$

then $\mathcal{R}_X = M \setminus X$.

Proof. $\mathcal{R}_X \subseteq M \setminus X$ as the balls are free. Fix $p \in M \setminus X$; if $p \in M_X$ then $p \in B(p, r(p)) \subseteq \mathcal{R}_X$. Otherwise ascend $d_X = d(\cdot, X)$ from p by its semiconcave generalised-gradient flow (Step 3 of Theorem 4.1): on the compact M the trajectory has finite length and converges to a critical point a_∞ of d_X , retaining p in the moving free ball, so $d(p, a_\infty) < r(a_\infty) = d_X(a_\infty)$. By δ -density $d_X \leq \delta$ on all of M , whence $r(a_\infty) \leq \delta < \frac{1}{2} \text{inj}(M)$; thus a_∞ lies off the cut locus of each of its nearest points, its ≥ 2 minimising geodesics (it is a critical point) reach ≥ 2 *distinct* points, and $a_\infty \in M_X$. Hence $p \in B(a_\infty, r(a_\infty)) \subseteq \mathcal{R}_X$. The density bound is what forces every critical point below the cut locus, turning the two-geodesic centre the flow finds into a two-*point* one. $\square$

Remark 6.2 (The capture theorem halves the constant and drops compactness). The final note of the series strengthens Theorem 6.1 on both hypotheses: on any *complete* M with $\text{inj}(M) := \inf_q \text{inj}(q) > 0$, already $\sup_{M \setminus X} d_X < \text{inj}(M)$ — for δ -dense X , $\delta < \text{inj}(M)$ — gives $\mathcal{R}_X = M \setminus X$ [5, Cor. 3.3]. Compactness entered only to preclude escape, which the spread bound does by itself; and the constant halves because the capture argument of [5] needs the reach bound only at the first multi-geodesic point it meets, not along a rival-forcing comparison of length $\sim 2\delta$.

Remark 6.3 (Positive curvature: making the hypothesis explicit). Theorem 6.1 is curvature-free, but a curvature upper bound turns it into an effective statement, which is the natural positive-curvature reading. By Klingenberg's lemma, for compact M ,

$$\text{inj}(M) \geq \min\left\{\frac{\pi}{\sqrt{\Delta}}, \frac{1}{2}\ell(M)\right\} \quad (\text{sec} \leq \Delta),$$

where $\ell(M)$ is the length of the shortest closed geodesic and $\frac{\pi}{\sqrt{\Delta}}$ is the conjugate radius bound (Rauch). Thus if $\kappa \leq \text{sec} \leq \Delta$ and X is δ -dense with

$$2\delta < \min\left\{\frac{\pi}{\sqrt{\Delta}}, \frac{1}{2}\ell(M)\right\},$$

then $\mathcal{R}_X = M \setminus X$. For $M = \mathbb{S}^n$ ($\kappa = \Delta = 1$, $\ell = 2\pi$) this gives total reconstruction once X is $\frac{\pi}{2}$ -dense; but Theorem 4.1 shows that on $\mathbb{S}^n$ the density hypothesis is in fact *unnecessary*. That is the content of the next remark: constant curvature is precisely where the spread condition can be dropped.

Remark 6.4 (Why constant curvature needs no spread). In Theorem 6.1 the spread hypothesis caps the reach, $d_X \leq \delta < \text{inj}(M)$, placing every critical point of d_X below the cut locus, where its ≥ 2 minimising geodesics reach *distinct* points. On $\mathbb{S}^n$ this is automatic for *any* X with $|X| \geq 2$, with no spread: the cut locus of each point is the single antipode at the *diameter* π , so a critical point (reach $< \pi$, as $|X| \geq 2$) already has two distinct nearest points — equivalently, along a nearest-point ray the antipode is where x_0 becomes the globally farthest contact, so a rival overtakes it first (Step 1 of Theorem 4.1). The two italicised features — the cut point realises the diameter, and it does so in every direction simultaneously — are exactly the rigidity of constant positive curvature. They fail in variable positive curvature, which is the subject of §7.

7 The variable-curvature obstruction: cut shielding

Off constant curvature, the conclusion $\mathcal{R}_X = M \setminus X$ can fail for sparse X . The obstruction is structural and worth stating precisely, even though we do not have a closed-form characterisation in the variable case: it is the only gap between Theorem 6.1 and the unconditional Theorem 4.1.

The proof of Theorem 4.1 produces a medial centre by following γ from x_0 until a *second contact* appears. If instead γ reaches the *cut locus* of x_0 first — a point $m = \gamma(\ell)$ admitting a second minimising geodesic back to the same x_0 , with no rival contact of X yet within distance ℓ — then m is not medial (x_0 is still its unique nearest *point* of X , merely reached twice), the margin identity breaks past m (there $d(\gamma(t), x_0) < t$), and the construction stalls. We call such a configuration *cut-shielding* of the points $\gamma(s)$, $s < \ell$.

Proposition 7.1 (The margin argument fails exactly at cut shielding). *Let $p \notin X$ have unique nearest point x_0 , and let $\ell = \text{cut}(x_0; \gamma)$ be the distance to the cut locus of x_0 along the ray γ through p . If no point of $X \setminus \{x_0\}$ lies within distance ℓ of any $\gamma(t)$, $t \leq \ell$ (i.e. x_0 is the unique nearest point of X all along $\gamma|_{[0, \ell]}$), then the moving-centre construction of Theorems 4.1–6.1 produces no medial centre on γ .*

Proof. For $t \leq \ell$ the arc is minimising and x_0 is the unique nearest point, so $d(\gamma(t), X) = t$ with $\text{Feet}(\gamma(t)) = \{x_0\}$: no $\gamma(t)$, $t \leq \ell$, is medial. At $t = \ell$ the point $m = \gamma(\ell)$ has a second minimising geodesic to x_0 , but its unique nearest *point* of X is still x_0 , so $m \notin M_X$. For $t > \ell$, $d(\gamma(t), x_0) < t$, so the identity $d(\gamma(t), X) = t$ fails and the constant margin $r - d(p, \cdot) = s_p$ is no longer maintained. $\square$

Remark 7.2 (A concrete model, and the honest status). The mechanism is visible on a prolate ellipsoid of revolution (a “cigar”, $\text{sec} > 0$ everywhere) with a short waist closed geodesic of length ℓ_0 . Take $X = \{x_0, x_1\}$ with x_0 on the waist and x_1 at the far tip. The waist geodesic through x_0 returns to x_0 along two arcs of length $\ell_0/2$; the diametrically opposite waist point m is a cut point of x_0 with x_0 still its unique nearest contact (the tip x_1 being far). By Proposition 7.1 the waist points between x_0 and m are not reconstructed by waist centres; whether some *off*-waist medial centre nonetheless reconstructs them is decided by the *global* two-point medial axis, which for $|X| = 2$ is exactly the bisector $\{a : d(a, x_0) = d(a, x_1)\}$, so $q \in \mathcal{R}_X$ iff some bisector point a has $d(a, q) < d(a, x_0)$. We have now settled this finite test numerically, and the answer is *negative*: cut-shielding genuinely obstructs two-point reconstruction once the curvature varies enough.

- *Unconditional totality*: constant curvature $+1$ (Theorem 4.1); and any compact M with δ -dense X , $2\delta < \text{inj } M$ (Theorem 6.1). These cover all *sufficiently sampled* X in positive curvature.
- *Settled (numerically, negative) for sparse X* : computing the bisector test above on a prolate ellipsoid of revolution (the cigar; $\text{sec} > 0$ everywhere), with the whole pipeline validated on the round sphere (geodesic distances within 1%; reconstruction total, zero shielded points, confirming Theorem 4.1), gives a sharp transition in the eccentricity. Reconstruction is total for small eccentricity and shielded for large, with the transition localised to $c^* = 2.6 \pm 0.1$ (the 2.4% solver error cannot resolve it more finely); the shielded region sits near the far tip. At $c = 3$ it is $\approx 9\%$ of the surface with reconstruction deficit $\min_a [d(a, q) - r(a)] \approx 18\%$ of the diameter at its deepest point (the distance by which q lies beyond every medial ball), a factor of about seven above the 2.4% discretisation error, and persisting and stabilising under mesh refinement (its area fraction rising slightly, not vanishing). So $\mathcal{R}_X = M \setminus X$ *fails* for sparse X in variable positive curvature: total two-point reconstruction is special to constant curvature (and, more finely, bounded eccentricity). The remaining open question is only the *quantitative* threshold — precisely, $\text{cut}(x_0; \gamma) \geq \sup\{t : x_0 \text{ uniquely nearest}\}$ along every ray, which on $\mathbb{S}^n$ holds because the cut distance π realises the diameter (Remark 6.4) and which the eccentricity controls on the cigar.

The shielded points are cut points of x_0 admitting two or more minimising geodesics to x_0 yet strictly closer to x_0 than to x_1 (a single nearest *point*, reached twice), hence lie in $M_X^g \setminus M_X$ and *are* reconstructed by the two-geodesic axis of §8 (Theorem 8.3): cut-shielding is the exact phenomenon separating the two axes off constant curvature, and the two-geodesic reformulation is its unconditional resolution.

Remark 7.3 (Constant-curvature space forms). For a spherical space form $\mathbb{S}^n/\Gamma$ ($\text{sec} \equiv +1$, compact) the local geometry is that of $\mathbb{S}^n$ but the cut locus of a point is no longer a single antipode (e.g. $\mathbb{RP}^n$ has diameter $\frac{\pi}{2}$ and cut locus a hyperplane $\mathbb{RP}^{n-1}$). Theorem 6.1 applies whenever X is δ -dense with $2\delta < \text{inj}(\mathbb{S}^n/\Gamma)$; for sparse X the cut locus is again the only obstruction, now genuinely nontrivial, and we leave the sharp statement open. Thus even within constant curvature it is the *simple connectivity* of $\mathbb{S}^n$ — forcing the cut locus to a single diameter-realising point — that makes total reconstruction unconditional.

8 The two-geodesic medial axis

The cut-shielding obstruction of §7 is an artefact of *which* medial axis one reconstructs from. Białożyty’s M_X counts points with two distinct nearest *points* of X . A shielded point $m = \gamma(\ell)$

has a single nearest point x_0 reached by two distinct minimising *geodesics* — it is invisible to M_X but is exactly the kind of point the cut locus is made of. Reconstructing instead from the *two-geodesic* medial axis removes the obstruction altogether, and does so without disturbing the sphere or the Euclidean theory.

Definition 8.1 (Two-geodesic medial axis). *For $a \in M \setminus X$ let $r(a) = d(a, X)$. Say $a \in M_X^g$ if there are two distinct minimising geodesics from a to X (their endpoints in X may coincide). Put $\mathcal{R}_X^g = \bigcup_{a \in M_X^g} B(a, r(a))$. Since two distinct nearest points give two distinct geodesics, $M_X \subseteq M_X^g$, hence $\mathcal{R}_X \subseteq \mathcal{R}_X^g \subseteq M \setminus X$; and M_X^g is (the two-sided part of) the singular set of $d(\cdot, X)$ — the cut locus of X .*

Proposition 8.2 (The change is invisible in $\mathbb{R}^n$ and on $\mathbb{S}^n$). *If $M = \mathbb{R}^n$, or $M = \mathbb{S}^n$ with $|X| \geq 2$, then $M_X^g = M_X$ and $\mathcal{R}_X^g = \mathcal{R}_X$.*

Proof. In $\mathbb{R}^n$ any two points are joined by a unique geodesic, so two distinct minimising geodesics to X have distinct endpoints, i.e. two nearest points. On $\mathbb{S}^n$, two distinct minimising geodesics from a to a *single* $x_0 \in X$ force a into the cut locus of x_0 , which is the antipode $\{-x_0\}$ at distance π ; then $r(a) = d(a, X) = \pi$ makes x_0 the unique point of X , contradicting $|X| \geq 2$. So in both cases distinct geodesics have distinct endpoints, $M_X^g = M_X$. $\square$

Thus Theorem 4.1 (total on $\mathbb{S}^n$) and Białożyty’s theorem (the defect structure in $\mathbb{R}^n$) are unchanged by the new definition. The definitions part company only where there is a cut locus — variable curvature, or nontrivial topology — which is exactly where the new one pays off.

Theorem 8.3 (Total reconstruction from the two-geodesic axis). *Let M be any compact Riemannian manifold. Then for every closed $X \subseteq M$ with $|X| \geq 2$,*

$$\mathcal{R}_X^g = M \setminus X.$$

No spread or curvature hypothesis is needed. The containment of p in the moving ball past intermediate cut points — the one formerly delicate step — is proved unconditionally in the final note of the series [5, Thm. 3.1, Cor. 3.2]; Remark 8.4 records the scope of the elementary margin argument and the repair.

Proof. $\mathcal{R}_X^g \subseteq M \setminus X$ as the balls are free. Fix $p \in M \setminus X$; if $p \in M_X^g$ then $p \in B(p, r(p)) \subseteq \mathcal{R}_X^g$. Otherwise ascend the semiconcave d_X from p (Step 3 of Theorem 4.1): on the compact M the trajectory has finite length and converges to a critical point a_∞ of d_X , retaining p in the moving free ball. A critical point has 0 in the convex hull of the unit initial directions of its minimising geodesics to X , hence *two or more* such geodesics — that is, $a_\infty \in M_X^g$, with no need to decide whether their endpoints coincide (which is exactly the focal/conjugate subtlety that the two-point axis must confront and the geodesic axis sidesteps). Thus $p \in B(a_\infty, r(a_\infty)) \subseteq \mathcal{R}_X^g$. $\square$

Remark 8.4 (Scope of the containment step). The clause “retaining p in the moving free ball” is Lieutier’s swallow property [6], proved in $\mathbb{R}^n$. In the elementary form of Step 3 the margin $r - d(p, \cdot)$ is constant along each unique-foot arc *because that arc is the minimising geodesic from its foot* — which holds exactly while the reach stays below the foot’s cut distance: unconditional on $\mathbb{S}^n$ (cut distance $\pi = \text{diameter}$) and under the spread hypothesis $2\delta < \text{inj } M$ of Theorem 6.1. On a general compact M the reach can exceed an intermediate foot’s cut distance, where the arc ceases to minimise and the constant-margin identity breaks. This gap is now closed: the final note of the series proves the containment unconditionally on every complete manifold, with no

genericity or analyticity hypothesis, by following the ascent only while the minimising geodesic to X is unique — there d_X grows at unit rate and the margin cannot shrink, through focal and conjugate points without comment — and *stopping at the first approach* to M_X^g , where the accumulated margin is intact, instead of flowing through it [5, Thm. 3.1, Cor. 3.2]. Theorem 8.3 therefore holds as stated for every compact M , and indeed for every closed $X \neq \emptyset$: even $|X| = 1$, invisible to the two-point axis, is reconstructed in full by the two-geodesic one.

Remark 8.5 (The fix activates exactly on the cut locus). The flow never needs a rival contact: when none appears it simply rests at a cut point of x_0 — two minimising geodesics to a single point — which the two-geodesic axis counts as medial. This is precisely the shielded point m of §7, now medial. The difference between the two theories,

$$M_X^g \setminus M_X = \{\text{points with a unique nearest point of } X \text{ but two minimising geodesics to it}\},$$

is empty in $\mathbb{R}^n$ and on $\mathbb{S}^n$ (Proposition 8.2) and carries the entire repair in variable curvature. So total reconstruction, which for M_X is special to constant curvature (Remark 6.4), holds for M_X^g on *every* compact manifold — positive curvature is no longer distinguished, because the mechanism is now compactness of the cut locus rather than the antipodal rigidity of the round metric.

Remark 8.6 (Which medial axis, and a caveat). The two axes answer different questions. M_X — points with two nearest *features* — is the symmetry/branching set, the shape skeleton of computational geometry; M_X^g — the singular set of $d(\cdot, X)$, i.e. the cut locus of X — is the natural object of the distance-function viewpoint [6] and of the Riemannian cut-locus theory. For the *reconstruction* question “is every point off X captured?” the geodesic axis is the clean choice, trivialising it on any compact manifold; for the *structural* question “what does the skeleton look like?” the point axis is the informative one, since M_X^g conflates a genuine branch of X with two routes to a single feature. Note that Theorem 8.3 needs no genericity: the gradient flow lands at a critical point, which carries two minimising geodesics by definition, so the focal/conjugate cut points that obstruct the two-point argument are absorbed without comment.

9 Discussion

The phenomenon, in one line. In non-positive curvature a point escapes reconstruction by sending its nearest-point ray to infinity with a single foot; medial reconstruction is therefore partial and is organised by a convexity defect of contact sets (Białożyty, [1, 2, 3, 4]). On the sphere geodesics are compact and the ray cannot escape: a second foot is forced before the antipode, every point is swallowed, and $\mathcal{R}_X = \mathbb{S}^n \setminus X$. Positive curvature thus removes the reconstruction obstruction *entirely*, the exact opposite of the expectation that a positively curved horosphere is the “hardest” case for detection — there is nothing left to detect.

Relation to the horospherical programme. In the Hadamard framework of [4] the ambient ideal boundary indexes the maximal free horoballs, and reconstruction outside the hull is drift to an ideal centre. On $\mathbb{S}^n$ the ideal boundary is *internalised*: the role of an ideal point is played by a genuine pole v , the Busemann function by the signed distance $d(\cdot, v) - \frac{\pi}{2}$ to the equator, and “drift to ξ ” by convergence of the moving centre $\gamma(t)$ to a medial point at finite distance. The curvature-free swallow lemma of [4] survives verbatim (Remark 4.2); what does *not* survive is the backbone that made the swallow’s hypothesis (a drifting medial centre) generic only at defective horoballs — on $\mathbb{S}^n$ every direction supplies one, so the swallow runs everywhere.

Where nontrivial positive-curvature structure lives. Total reconstruction makes the *set* reconstruction problem trivial on the sphere, which suggests the informative positive-curvature question is the *domain* reconstruction problem of Lieutier [6]: for an open $\Omega \subseteq \mathbb{S}^n$, reconstruct Ω from the medial axis of $\partial\Omega$ using only balls *contained in* Ω . There the focal centres on the far side of $\partial\Omega$ are excluded by the confinement to Ω , the antipodal collapse does not occur, and a genuine convexity/defect theory — now with the spherical, positively curved induced geometry of $\partial\Omega$ — should re-emerge. That is the natural sequel.

Summary.

Statement	Setting	Status
Focal pole: $ K_v \geq 2 \Rightarrow H_v^\circ \subseteq \mathcal{R}_X$	$\mathbb{S}^n$	Lem. 3.1
Total reconstruction $\mathcal{R}_X = \mathbb{S}^n \setminus X$	$\mathbb{S}^n$, any closed X , $ X \geq 2$	Thm. 4.1
Białozyt certificate strictly undercounts	$\mathbb{S}^n$ (cap, circle)	Prop. 5.1, 5.2
Total reconstruction under spread $2\delta < \text{inj } M$	any compact M	Thm. 6.1
Explicit via $\text{sec} \leq \Delta$ (Klingenberg)	compact, $\text{sec} \leq \Delta$	Rem. 6.3
Cut-shielding obstruction; sparse variable case	compact $\text{sec} > 0$	open (Prop. 7.1)
Two-geodesic axis $\mathcal{R}_X^g = M \setminus X$ (no spread)	any compact M	Thm. 8.3 , [5]
$M_X^g = M_X$ (definition invisible)	$\mathbb{R}^n, \mathbb{S}^n$	Prop. 8.2

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